

Board Games Dataset (BGD) Project Report

Data Mining: Fundamentals

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1. Data understanding and preparation

1.1 Dataset overview

The dataset consists of 21,925 records, each representing a board game collected from BoardGameGeek (BGG), a large online board gaming community. Each instance is described by 46 attributes, whose definitions, semantic grouping, and data types are summarized in Table 1.1.

TABLE 1.1: Dataset attributes overview, grouped by semantic category.

Attribute	Description	Type
<i>Identifiers and Metadata</i>		
BGGId	Unique game identifier	Identifier
Name	Game title	Text
Description	Textual game description	Text
ImagePath	URL to game image	Text
YearPublished	Publication year	Discrete
<i>Gameplay Specifications</i>		
MinPlayers	Minimum number of players	Discrete
MaxPlayers	Maximum number of players	Discrete
BestPlayers	Community-voted optimal player count	Discrete
GoodPlayers	Community-voted suitable player counts	Discrete (multi-valued)
MfgPlaytime	Manufacturer-stated play time (min)	Discrete
ComMinPlaytime	Community minimum play time (min)	Discrete
ComMaxPlaytime	Community maximum play time (min)	Discrete
MfgAgeRec	Manufacturer age recommendation	Discrete
ComAgeRec	Community age recommendation	Continuous
<i>Complexity Metrics</i>		
GameWeight	Game complexity (1–5 scale)	Continuous
ComWeight	Community-rated complexity (1–5 scale)	Continuous
LanguageEase	Language dependency score	Continuous
<i>Community Engagement</i>		
NumOwned	Number of owners	Discrete
NumWant	Number of users wanting the game	Discrete
NumWish	Number of wishlist additions	Discrete
NumUserRatings	Number of user ratings	Discrete
NumComments	Number of user comments	Discrete
NumWeightVotes	Votes for complexity rating	Discrete
<i>Game Variants</i>		
NumAlternates	Number of alternate versions	Discrete
NumExpansions	Number of expansions	Discrete
NumImplementations	Number of implementations	Discrete
IsReimplementation	Reimplementation flag	Binary

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Attribute	Description	Type
<i>Publishing and Funding</i>		
Family	Editorial game family or series	Categorical
Kickstarted	Crowdfunding origin flag	Binary
<i>Category Rankings (Rank:*)</i>		
Rank:strategygames	Rank in strategy category	Discrete
Rank:familygames	Rank in family category	Discrete
Rank:thematic	Rank in thematic category	Discrete
Rank:wargames	Rank in war games category	Discrete
Rank:partygames	Rank in party category	Discrete
Rank:childrensgames	Rank in children's category	Discrete
Rank:abstracts	Rank in abstract category	Discrete
Rank:cgs	Rank in card games category	Discrete
<i>Category Membership (Cat:*)</i>		
Cat:Strategy	Strategy category membership	Binary
Cat:Family	Family category membership	Binary
Cat:Thematic	Thematic category membership	Binary
Cat:War	War category membership	Binary
Cat:Party	Party category membership	Binary
Cat:Childrens	Children's category membership	Binary
Cat:Abstract	Abstract category membership	Binary
Cat:CGS	Card games category membership	Binary
<i>Game Evaluation</i>		
Rating	Game rating	Ordinal

1.2 Data quality assessment and treatment

1.2.1 Missing values analysis

Pairwise correlation analysis of missingness indicators revealed predominantly independent patterns, with no evidence of systematic co-occurrence across attributes.

BestPlayers and *GoodPlayers* contained placeholder values (0 and empty lists, respectively) indicating absence of community votes rather than true data. With 90.90% of records affected simultaneously, imputation was infeasible, and both features were removed.

Family exhibited severe data quality limitations: 69.61% missingness, 1,456 distinct categories with an average frequency below 5 occurrences each, and no category exceeding 0.47% coverage. Due to extreme sparsity and absence of meaningful structure, this feature was removed.

ComAgeRec, with a missing rate of 25.22%, showed significant variation across game categories. A category-based median imputation strategy was adopted: for each record with missing values, the median was computed from the category-specific medians of its active *Cat:** attributes, with a global median (10.0) as fallback.

GameWeight contained 506 zero values (2.31%) on a defined [1, 5] scale, indicating games with no community votes. These entries were treated as missing and imputed using the same category-based median strategy, with a global median of 2.0 as fallback.

1.2.2 Semantic inconsistencies and non-analytic features

LanguageEase was documented as a language dependency indicator but exhibited an anomalous value range (1–1,757) with high variance ($\sigma = 236.6$) and 26.87% missing entries. Correlation analysis confirmed no meaningful relationship with other features ($|\rho| < 0.27$ for all). Given the lack of interpretable structure, the feature was removed.

BGGId, *ImagePath*, and *Description* do not provide structured information for the current analysis: *BGGId* is a unique identifier, *ImagePath* contains URLs, and *Description* consists of unstructured text. All three were excluded from the analytical pipeline.

1.3 Feature correlation and redundancy reduction

To identify and eliminate multicollinearity, pairwise Spearman rank correlation analysis was performed on all numerical features. Spearman’s coefficient was selected due to its robustness to non-normality and its appropriateness for ordinal and bounded variables.

1.3.1 Highly correlated feature pairs

Figure 1.1 presents the Spearman correlation matrix for features exhibiting high pairwise correlations.

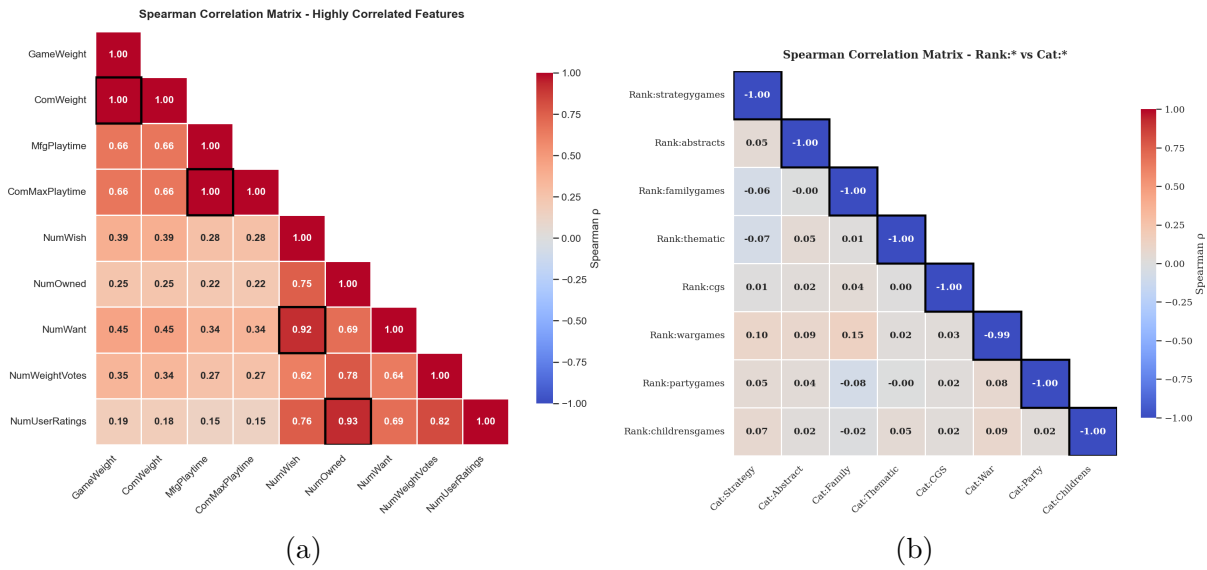


FIGURE 1.1: Spearman correlation matrices for features exhibiting high pairwise correlations: (a) selected numerical features; (b) cross-correlation between *Rank:** and *Cat:** features. Feature pairs with $|\rho| \geq 0.90$ are outlined in black.

GameWeight and *ComWeight* both represent game complexity on a bounded scale [1, 5] and exhibit perfect correlation ($\rho = 1.00$). *ComWeight* was removed because it contained 13 out-of-range values (> 5), whereas *GameWeight* contained only semantic zeros, treated as missing and imputed using the category-based strategy described in Section 1.2.1.

MfgPlaytime and *ComMaxPlaytime* were found to be perfect duplicates ($\rho = 1.00$), sharing identical values and missingness patterns. *MfgPlaytime* was therefore removed.

Despite high correlation, *NumOwned* and *NumUserRatings* ($\rho = 0.93$) were both retained as they capture semantically distinct constructs: ownership versus evaluation activity.

NumOwned, *NumWant*, and *NumWish* exhibited strong inter-correlations ($\rho = 0.69$ – 0.92), with *NumWant* and *NumWish* showing the highest redundancy ($\rho = 0.92$). As these features measure

closely related aspects of user interest, they were consolidated into a single *InterestScore* metric (see Section 1.3.2).

The *Rank:** features exhibited perfect negative correlations ($\rho = -1.0$) with the corresponding *Cat:** binary indicators, since rankings are assigned only to games belonging to a given category. All *Rank:** features were removed, retaining the binary *Cat:** indicators as a cleaner representation of category membership.

1.3.2 Feature aggregation

For feature groups exhibiting high inter-correlation and semantic overlap, aggregation was preferred over single-feature retention to preserve information while reducing dimensionality.

PlaytimeComAvg was computed as the arithmetic mean of *ComMinPlaytime* and *ComMaxPlaytime*, providing a single estimate of typical game duration while mitigating the effect of extreme playtime ranges.

InterestScore consolidates *NumOwned*, *NumWant*, and *NumWish* into a unified measure (arithmetic mean), reducing dimensionality while preserving the semantic content of the original features.

1.4 Data transformation and outlier handling

1.4.1 Variance-stabilizing transformations

Several numerical features exhibited severe positive skewness (ranging from 11.51 to 84.64). To mitigate the impact of heavy-tailed distributions, both logarithmic and square root transformations were evaluated (Table 1.2).

TABLE 1.2: Comparison of skewness (Skew) and kurtosis (Kurt) before and after logarithmic and square root transformations. Bold values indicate the selected transformation for each feature.

Feature	Original		Log		Sqrt		Selected
	Skew	Kurt	Skew	Kurt	Skew	Kurt	
<i>PlaytimeComAvg</i>	17.70	514.94	0.62	1.20	4.10	36.34	Log
<i>InterestScore</i>	11.51	197.38	0.84	0.62	4.03	24.89	Log
<i>NumUserRatings</i>	12.67	233.35	1.10	0.91	4.57	30.93	Log
<i>NumWeightVotes</i>	15.49	371.48	0.86	0.92	4.55	33.44	Log
<i>NumAlternates</i>	84.64	8857.49	1.47	1.96	3.31	52.87	Log
<i>NumExpansions</i>	25.54	1266.85	2.63	7.90	4.27	32.68	Log
<i>NumImplementations</i>	12.23	370.65	2.01	4.65	1.97	4.92	Log

The logarithmic transformation consistently outperformed the square root alternative and was therefore applied to all seven features. Figure 1.2 illustrates the effect of each transformation on *PlaytimeComAvg*.

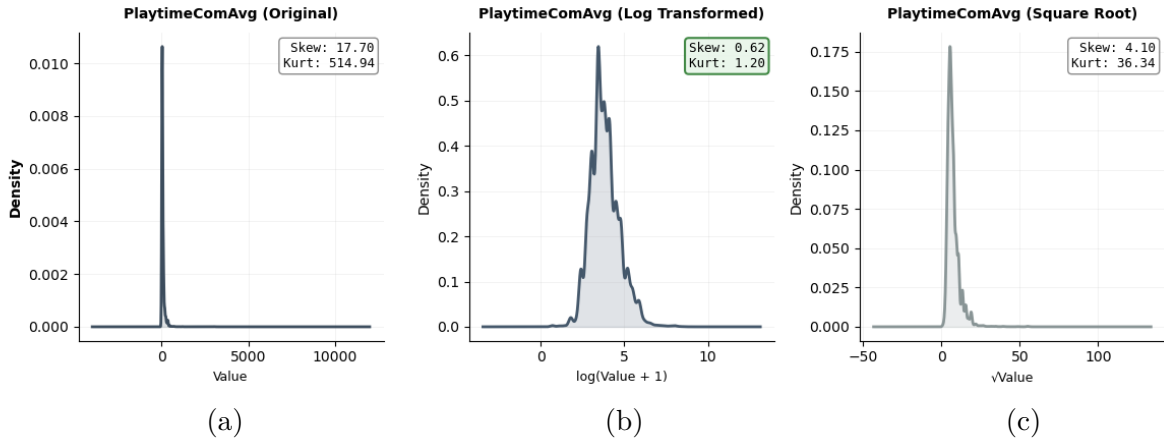


FIGURE 1.2: Effect of different transformations on the distribution of *PlaytimeComAvg*: (a) original scale; (b) log-transformed; (c) square-root transformed. Skewness and kurtosis values are reported for each case.

1.4.2 Label encoding

The *Rating* variable is an ordinal categorical feature with three distinct levels: *Low*, *Medium*, and *High*. The distribution is dominated by *Medium*-rated observations (43.99%), followed by *Low* (33.04%) and *High* (22.97%).

Given the inherent hierarchy, the variable was encoded using an integer ordinal scale (*Low* = 1, *Medium* = 2, *High* = 3), preserving the semantic ranking while avoiding the unnecessary dimensionality of one-hot encoding.

1.4.3 Temporal filtering

A cleaning step was applied to *YearPublished* to ensure temporal coherence. A total of 1.85% of records reported publication years prior to 1950, including entries with year 0 (0.88%, interpreted as placeholders for missing data) and extremely early dates, in some cases dating back to 3500 B.C.

A threshold of 1950 was introduced to focus on modern board game publishing trends while removing potential data entry errors and historically rare samples.

1.4.4 Outlier detection and removal

A feature-specific strategy was adopted for outlier detection based on distributional characteristics. *MaxPlayers*, exhibiting extreme right-skewness (42.66) and heavy tails (kurtosis 2657.50), was the only feature for which outlier removal was applied.

An interquartile range (IQR) method with a conservative multiplier of 4 was employed (i.e., values exceeding $Q_3 + 4 \times \text{IQR}$ were flagged as outliers). This procedure removed 454 records (2.11% of the dataset), yielding a substantially more regular distribution (Figure 1.3).

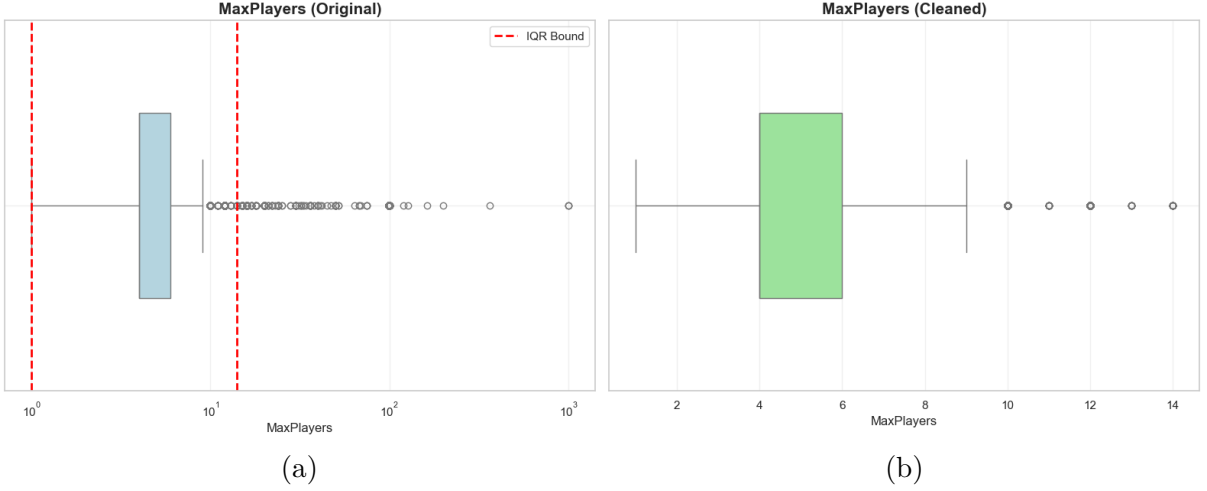


FIGURE 1.3: Boxplots of *MaxPlayers* before and after outlier removal. (a) Original distribution (log scale) showing extreme right-skewness (Skew = 42.66, Kurt = 2657.50). (b) Distribution after IQR-based filtering ($k = 4$), with improved shape (Skew = 0.89, Kurt = 1.53).

2. Clustering

2.1 K-means

A subset of features (*GameWeight*, *PlaytimeComAvg*, *MinPlayers*, *NumUserRatings*) was selected for K-means clustering and scaled using *MinMaxScaler*. Principal Component Analysis (PCA) confirmed the soundness of this selection: the first two components jointly explain 91.25% of the total variability, indicating an almost two-dimensional space (Figure 2.1), particularly suitable for Euclidean distance.

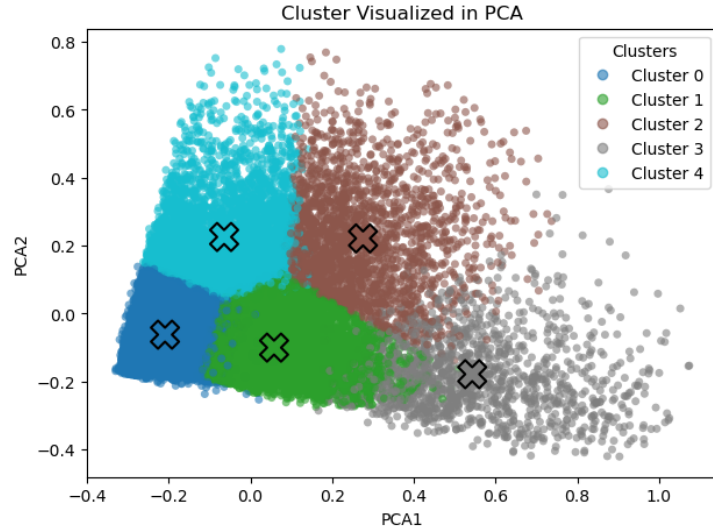


FIGURE 2.1: K-means clustering visualized in the space of the first two principal components (PC1 and PC2). Cluster centroids are marked with crosses.

Clustering was evaluated for $k \in [2, 29]$. The Sum of Squared Errors (SSE) shows a smooth progressive decrease without discontinuities (Figure 2.2a). Silhouette scores reach their maximum

at $k = 3$ (0.375), but values for k between 4 and 6 remain fairly stable (Figure 2.2b). The choice of $k = 5$ represents a compromise between separation quality and cluster interpretability.

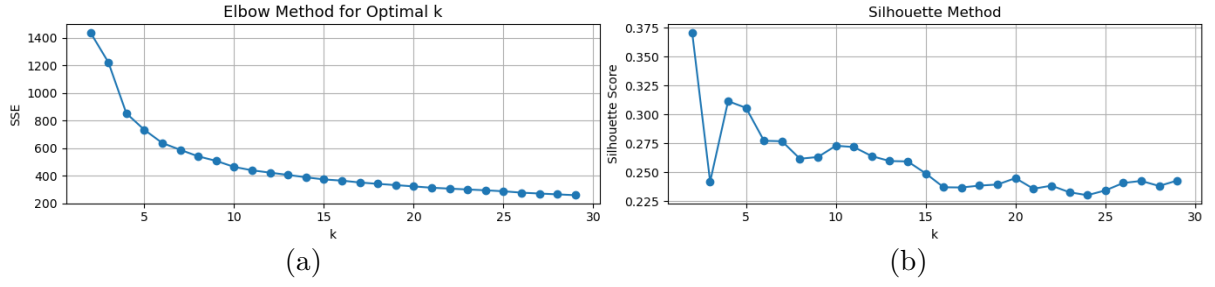


FIGURE 2.2: K-means evaluation metrics for $k \in [2, 29]$. (a) Sum of Squared Errors (SSE). (b) Silhouette score.

The final segmentation ($k = 5$, $\text{SSE} = 501$, $\text{Silhouette} = 0.312$) is well balanced, with cluster sizes ranging from 10.91% to 28.74%, indicating dense and well-distributed regions without dominant clusters.

2.2 DBSCAN

DBSCAN was applied to a subset of six numerical features: *GameWeight*, *ComAgeRec*, *MinPlayers*, *YearPublished*, *PlaytimeComAvg*, and *NumUserRatings*, normalized using *MinMaxScaler*.

The k -th nearest neighbor distance plot was used to guide parameter selection (Figure 2.3). Distances were computed for $k = 5, 10, 25, 50, 100, 200$, suggesting a candidate ε range between 0.02 and 0.15. The distance curves exhibit a gradual inflection rather than a sharp elbow, indicating a gradual density structure without well-defined density thresholds.

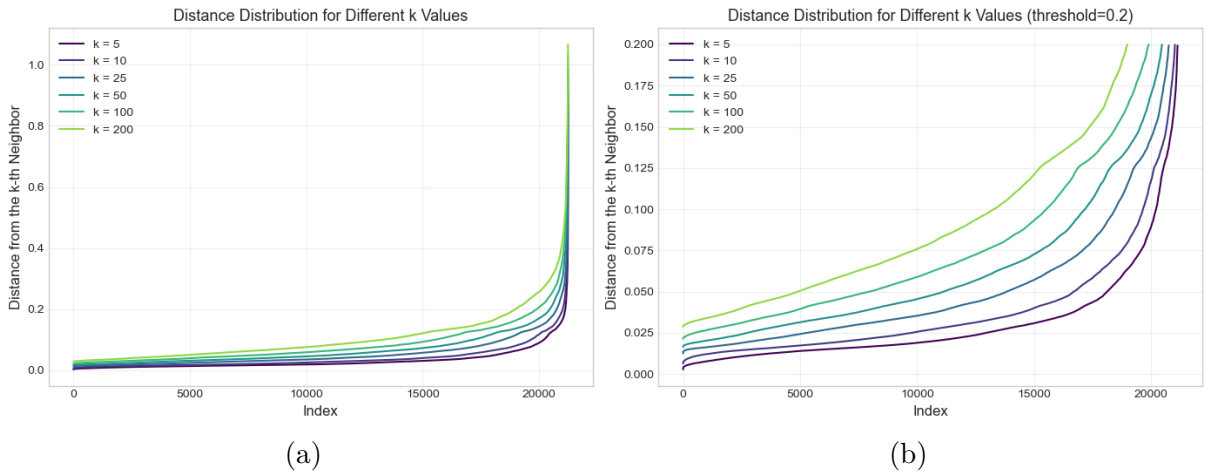


FIGURE 2.3: K-th nearest neighbor distance analysis. (a) Full visualization showing the distance distribution for different k values. (b) Zoomed visualization with threshold = 0.2 to identify potential elbow points.

An exhaustive grid search was then conducted to examine DBSCAN behavior across the parameter space. The search varied ε from 0.02 to 0.15 (in steps of 0.01) and *MinPts* from 5 to 50 (in steps of 5), yielding 140 configurations. For each configuration, descriptive indicators were recorded, including the number of clusters, noise ratio, and silhouette score (Figure 2.4).

The number of clusters decreases as ε increases. For *MinPts* = 5, over-fragmentation is observed at low ε values (over 200 clusters at $\varepsilon = 0.02$), while higher *MinPts* values (15, 30, 50) produce fewer clusters throughout the range with more stable behavior.

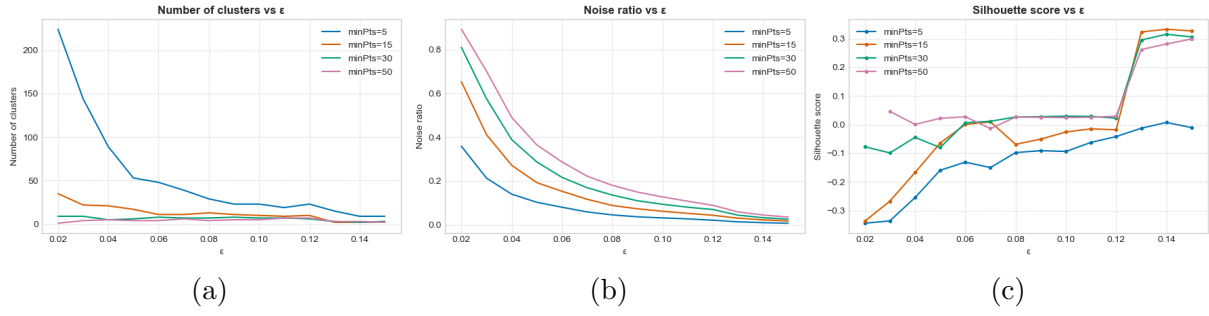


FIGURE 2.4: DBSCAN parameter sensitivity analysis. (a) Number of detected clusters as a function of ε . (b) Noise ratio as a function of ε . (c) Silhouette score as a function of ε , for different MinPts values.

The noise ratio shows an inverse relationship with ε . At low ε , noise levels are high and strongly dependent on MinPts, while they progressively decline and converge as ε increases, falling below 10% for all configurations when $\varepsilon > 0.12$.

Silhouette scores generally increase with ε . At low ε values, scores are predominantly negative, indicating poor cluster separation. As ε increases, scores approach zero and become moderately positive for some MinPts values beyond $\varepsilon \approx 0.12$. However, this apparent improvement coincides with a strong reduction in the number of clusters and reflects trivial aggregation rather than meaningful cluster structure.

The configuration $\varepsilon = 0.15$ and MinPts = 15 was further inspected. Although this setting produces a low noise level (1.7%) and a silhouette score of 0.32, the resulting cluster composition is highly imbalanced, with a single dominant cluster containing 95.0% of the data. This outcome does not represent a meaningful segmentation.

Additional experiments using alternative feature subsets did not improve results. This behavior reflects the dataset's dense and uniform structure: the absence of natural density gaps prevents DBSCAN from identifying well-separated regions. The silhouette score, biased toward globular clusters, proved unsuitable for evaluating density-based methods, as positive values at high ε merely reflect trivial aggregation rather than meaningful structure. For these reasons, DBSCAN is not retained as a final clustering solution.

2.3 Hierarchical clustering

Hierarchical clustering was performed in two stages. First, it was applied to a subset of features not yet considered in previous analyses. Subsequently, Ward's method was compared with K-means using the same features as in Section 2.1, to assess the robustness of the partitional solution.

2.3.1 Hierarchical clustering on complementary feature space

Hierarchical clustering was applied to features not included in the centroid-based approach: *MaxPlayers*, *ComAgeRec*, *InterestScore*, *NumWeightVotes*, *NumAlternates*, *NumExpansions*, and *NumImplementations*, representing scalability, accessibility, community engagement, and product ecosystem attributes.

As shown in Figure 2.5, most linkage methods exhibit significant limitations, producing highly imbalanced partitions with dominant clusters. Only Ward's method yields partially acceptable results, with cluster balance improving substantially from $k \geq 5$.

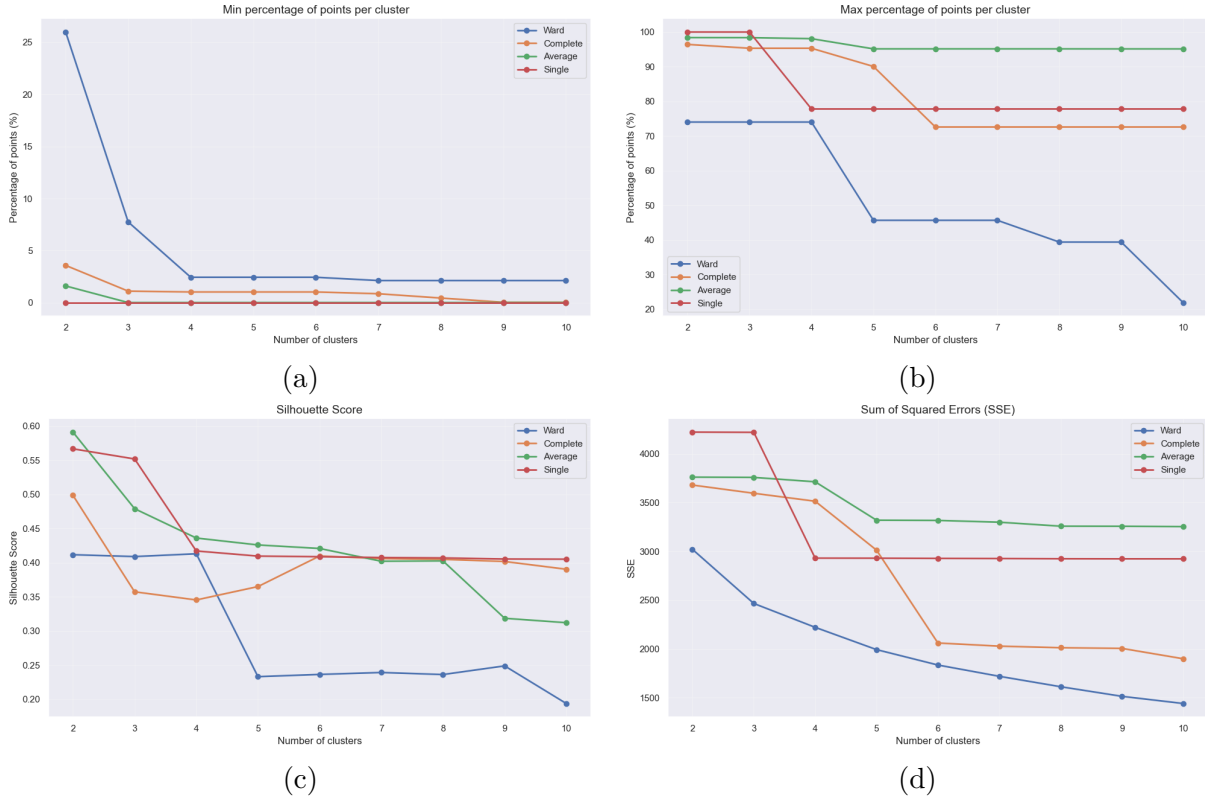


FIGURE 2.5: Hierarchical clustering evaluation across linkage methods for $k \in [2, 10]$. (a) Minimum cluster size (%). (b) Maximum cluster size (%): Single linkage assigns 100% for $k = 2-3$, 77.78% for $k \geq 4$; Average linkage $>95\%$ for $k = 2-10$; Complete linkage similar for $k = 2-5$; Ward decreases from 74.01% ($k \leq 4$) to 45.65% ($k = 5$) and 21.87% ($k = 10$). (c) Silhouette score. (d) Sum of Squared Errors (SSE).

These behaviors stem from an extremely high-density core combined with high dimensionality. Ward's dendrogram (Figure 2.6) confirms these challenges, revealing heterogeneous merge distances ($\sim 14-20$ for cohesive sub-clusters to ~ 48 for peripheral clusters). For $k = 5$, despite improved balance (maximum cluster size 45.65%), uneven spacing between merge points indicates limited separation — a limitation that will be addressed in Section 2.3.2 with better feature selection.

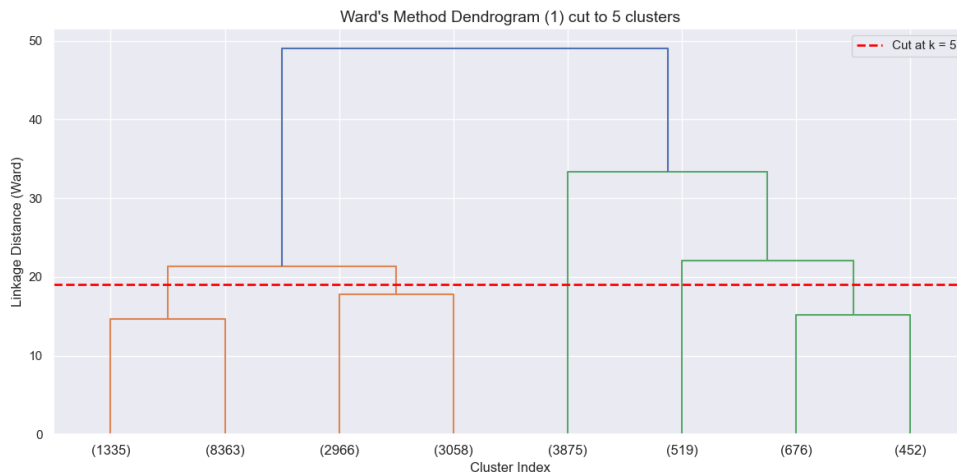


FIGURE 2.6: Ward's method dendrogram with remaining features. The horizontal dashed line indicates the cut at $k = 5$, producing five clusters of unequal sizes. Cluster sizes (number of observations) are reported at the bottom of each branch.

2.3.2 Comparison between K-means and Ward's method

In the second stage, Ward's hierarchical clustering was applied to the same subset of features and with the same number of clusters ($k = 5$). Table 2.1 reports a direct comparison between K-means and Ward's method under identical feature conditions.

TABLE 2.1: Comparison of K-means and Ward's method using the same selected features ($k = 5$). Δ indicates the relative difference (K-means vs. Ward). K-means consistently outperforms Ward in separation (Silhouette: 0.312 vs 0.274), compactness (SSE: 501 vs 609), and balance (Max%: 28.7% vs 42.51%), confirming the robustness of the partitional solution.

Method	Silhouette	SSE	Max %	Min %
K-means	0.312	501	28.7%	10.9%
Ward	0.274	609	42.51%	4.82%
Δ	+13.9%	-18%	-32.5%	+126%

The comparison in Table 2.1 shows that Ward's performance improves substantially when using K-means features (Silhouette 0.274, SSE 609) versus complementary features (Silhouette 0.233, SSE 1994), indicating that feature selection significantly impacts hierarchical clustering quality.

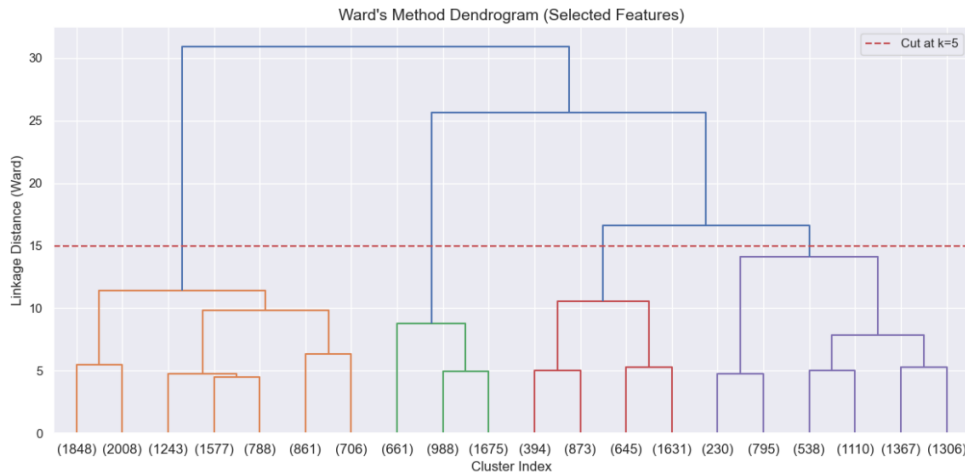


FIGURE 2.7: Ward's method dendrogram using the same features of K-means. The horizontal dashed line indicates the cut at $k = 5$. The dendrogram displays a more balanced structure compared to the dendrogram built on the complementary features. For $k = 5$, no cluster exceeds 42.51% of the observations. Nevertheless, K-means still delivers superior performance (Silhouette = 0.312, maximum cluster size = 28.7%), indicating that the intrinsic structure of the dataset is more favorable to partitional clustering approaches.

2.4 Summary

This chapter evaluated the effectiveness of partitional, density-based, and hierarchical clustering approaches in capturing the structure of the dataset. The comparative analysis shows that clustering outcomes are strongly influenced by both the intrinsic geometry of the data and the compatibility between this geometry and the assumptions underlying each method.

The results indicate that a centroid-based strategy represents the most appropriate choice. After feature selection and normalization, the data lie in a low-dimensional space in which distances are informative and cluster shapes are approximately globular. In this context, the K-means solution

with $k = 5$ achieves a satisfactory balance between compactness, separation, and interpretability, producing clusters of comparable size without dominance effects.

Density-based clustering fails to provide a meaningful partition. Across a wide range of parameter configurations, DBSCAN tends to fragment the data into numerous small components or to collapse them into a single large group. Although silhouette values increase in some cases, this behavior reflects a loss of structure rather than a genuine improvement in clustering quality.

Hierarchical clustering further confirms the importance of feature selection. Although Ward’s method benefits from the use of the same feature space adopted for K-means, it proves less effective in terms of cluster balance and internal cohesion, highlighting the limitations of hierarchical approaches for this dataset.

The analysis suggests that the data are organized into a small number of compact and moderately separated groups. Under these conditions, K-means with $k = 5$ represents the most coherent and robust clustering solution.

3. Classification

Classification was performed on the training set using three algorithms: K-NN (K-Nearest Neighbors), Naïve Bayes, and Decision Tree. Both a multiclass and a binary classification task were conducted, using *Rating* and *HighInteraction*¹ as target variables, respectively.

3.1 Multiclass classification

For the multiclass classification task, several alternative feature subsets were evaluated during model development. The final subset was selected as it provided a good trade-off between predictive performance and interpretability, without introducing overfitting. A domain-driven exploratory feature selection was then adopted, aiming to retain a heterogeneous set of features capturing multiple dimensions of board game characteristics. Accordingly, **GameWeight**, **YearPublished**, **ComAgeRec**, **InterestScore**, **MaxPlayers**, **NumExpansions**, and the **Strategy** and **War** category indicators were included.

3.1.1 K-Nearest Neighbours (KNN) - Multiclass classification

Given the algorithm’s sensitivity to feature scaling, numerical features were normalized using **StandardScaler** while binary categorical features were left unscaled, as they already exist on a comparable scale. The feature set was deliberately kept compact to avoid the curse of dimensionality, which degrades K-NN performance in high-dimensional spaces. The algorithm was optimized using 5-fold-cross-validation, with the best configuration achieved at $k = 20$, Manhattan distance, and uniform weighting.

The model achieved a test accuracy of 0.656, macro F1-score of 0.658, and minimal overfitting (gap: 2.27%), demonstrating strong generalization. Per-class metrics show precision of 0.67, 0.62, and 0.72 for *Low*, *Medium*, and *High* respectively, with corresponding recall values of 0.70, 0.66, and 0.59.

As shown in the confusion matrix (Table 3.1), substantial confusion occurs between adjacent classes: 28.4% of *Low*-rated games are misclassified as *Medium*, and 37.8% of *High*-rated games as *Medium*. Conversely, direct *Low-High* misclassifications remain rare (<3%), indicating effective discrimination between extremes. The asymmetric pattern, with *High* games more frequently

¹*HighInteraction* is a binary target derived from *NumUserRatings* by thresholding at the median value.

confused with *Medium* than *Low* games, suggests greater feature overlap at the upper rating boundary.

TABLE 3.1: Confusion matrix for K-NN (multiclass classification). Rows represent true labels; columns represent predicted labels.

	Predicted: Low (1)	Predicted: Medium (2)	Predicted: High (3)
True: Low	961	391	26
True: Medium	439	1246	197
True: High	31	374	584

The ROC curves (Figure 3.1) further illustrate the discriminative ability of the classifier across the three rating classes.

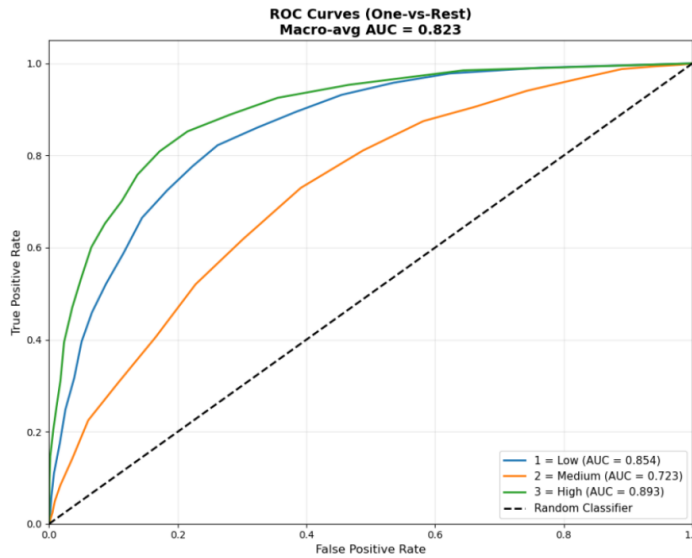


FIGURE 3.1: ROC curves for K-NN multiclass classification. Each curve represents the one-vs-rest performance for Low, Medium, and High rating classes. AUC values are reported in the legend.

3.1.2 Naïve Bayes classifier - Multiclass classification

For the Naïve Bayes, two variants of the algorithm were evaluated: Categorical and Gaussian Naïve Bayes. However, the heterogeneous feature set, comprising log-transformed continuous variables, low-cardinality numerical attributes, and binary categorical indicators, made Gaussian Naïve Bayes less suitable for the task, resulting in lower performance (accuracy: 0.53, weighted F1: 0.51). Consequently, continuous features were discretized into bins to leverage the Categorical Naive Bayes variant.

This approach shows varied performance across classes: *Low* (F1: 0.66, recall: 0.70) and *High* (F1: 0.62, recall: 0.62) show robust results, while *Medium* proves most challenging (F1: 0.57, recall: 0.54) despite being the most frequent class. As shown in the confusion matrix (Table 3.2), *Medium* observations are frequently misclassified as *Low* or *High*, reflecting significant overlap between adjacent classes.

TABLE 3.2: Confusion matrix for Categorical Naïve Bayes (multiclass classification). Rows represent true labels; columns represent predicted labels.

	Predicted: Low (1)	Predicted: Medium (2)	Predicted: High (3)
True: Low (1)	963	369	46
True: Medium (2)	531	1016	335
True: High (3)	58	313	618

The ROC curves (Figure 3.2) confirm this pattern, with *Low* and *High* classes achieving higher AUC values than *Medium*.

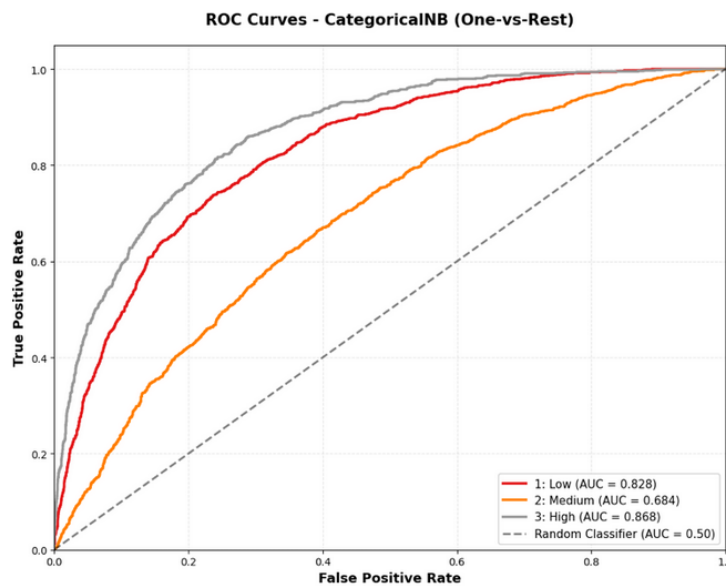


FIGURE 3.2: ROC curves for Categorical Naïve Bayes multiclass classification (one-vs-rest). AUC values are reported in the legend.

3.1.3 Decision Tree - Multiclass classification

The Decision Tree was implemented on the non-normalized dataset, as tree-based algorithms are scale-invariant. Two optimization strategies were employed: Grid Search and Randomized Search, both using 5-fold cross-validation.

Grid Search achieved a test accuracy of 0.6489, while Randomized Search identified a superior configuration with accuracy of 0.6507: Gini criterion, $\text{max_depth} = 10$, $\alpha_{ccp} = 0.000346$, $\text{min_samples_leaf} = 16$, and $\text{min_samples_split} = 26$. The more restrictive tree depth and stricter leaf constraints effectively balance bias and variance, preventing overfitting. Table 3.3 presents per-class performance metrics.

TABLE 3.3: Performance metrics for the optimized Decision Tree model (Randomized Search).

Class	Precision	Recall	F1-Score
Low (1)	0.68	0.65	0.67
Medium (2)	0.60	0.69	0.64
High (3)	0.73	0.58	0.65
Accuracy		0.6507	
Macro Avg	0.67	0.64	0.65
Weighted Avg	0.66	0.65	0.65

Despite careful parameter optimization, performance plateaued near 65% due to dataset limitations:

- Feature overlap: *GameWeight* and *YearPublished* show significant distribution overlap between *Medium* and *High* classes, preventing clean decision boundaries.
- Class imbalance: *Medium* dominates 44.2% of training data, leading to high recall (0.69) but lower precision (0.60).
- Pruning sensitivity: The specific α value (0.000346) was crucial to prevent overfitting or collapse of the minority *High* class.

Figure 3.3 shows the optimized tree structure. *GameWeight* (importance: 0.35) is the most important feature at the root node (threshold ≤ 1.847), followed by *YearPublished* (0.34), appearing across multiple levels. Secondary features include *NumUserRatings* (0.10) and *InterestScore* (0.08). Low-complexity games, especially older ones (pre-2010.5), predominantly receive Low ratings, while recent high-complexity games (post-2017.5, *GameWeight* > 2.005) tend toward High ratings.

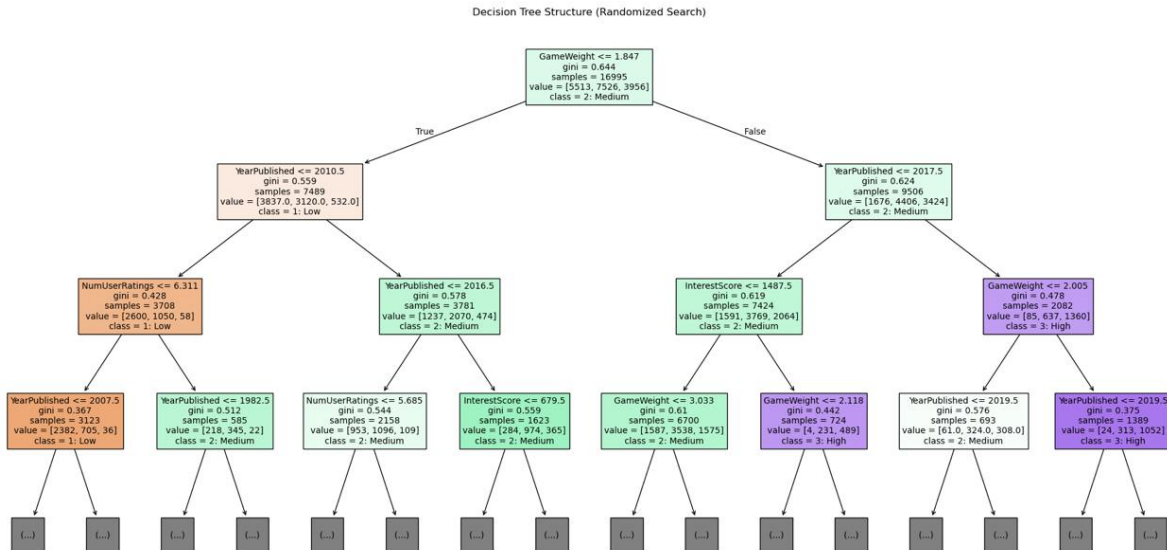


FIGURE 3.3: Optimized Decision Tree structure obtained through Randomized Search, showing the hierarchical split conditions and feature importance.

3.1.4 Summary

Overall, the multiclass classification task proved non-trivial, with K-NN achieving the best performance (accuracy: 0.656, macro F1: 0.658), followed by Decision Tree (0.6507) and Categorical Naïve Bayes (0.61). The consistent performance plateau around 65% across all models, despite rigorous optimization, suggests fundamental data-driven limitations rather than algorithmic constraints.

3.2 Binary classification

To complement the multiclass task, a binary classification was implemented using *HighInteraction* as target, identifying games whose *NumUserRatings* exceeds the median (4.828). *NumUserRatings* was excluded to prevent data leakage, while *NumWeightVotes* and *InterestScore* were removed due to strong correlation with user engagement (0.88 and 0.75 respectively). Low-predictive features (*PlaytimeComAvg*, *MinPlayers*, *ComAgeRec*) were also discarded.

3.2.1 K-Nearest Neighbours (KNN) - Binary classification

Numerical features were standardized using *StandardScaler* to address the algorithm’s sensitivity to feature scale, while binary features were left in their original form. Hyperparameter tuning via stratified 5-fold cross-validation identified the optimal configuration: $k = 15$, *uniform weights*, *Manhattan distance*. The model achieves test accuracy of 0.744 and AUC of 0.818, with macro F1-score of 0.74.

However, a performance discrepancy is observed: high recall for *Low Interaction* (0.86) but lower recall for *High Interaction* (0.63). This suggests the classifier is conservative in predicting the positive class, yielding high precision (0.82) but missing 37% of actual high-interaction games. The confusion matrix (Table 3.4) reveals this tendency, with the model more frequently misclassifying High Interaction instances as Low Interaction.

TABLE 3.4: Confusion matrix for K-NN (binary classification). Rows represent true labels; columns represent predicted labels.

	Predicted: Low (0)	Predicted: High (1)
True: Low (0)	1817	305
True: High (1)	783	1344

3.2.2 Naïve Bayes classifier - Binary classification

Both Categorical and Gaussian Naïve Bayes variants were evaluated, with the Categorical variant slightly outperforming the Gaussian one (accuracy: 0.76 vs 0.74, weighted F1: 0.76 vs 0.73).

The Categorical variant achieved accuracy of 0.761, macro-precision of 0.77, recall of 0.76, F1-score of 0.76, and AUC of 0.83. The confusion matrix (Table 3.5) provides further insight into the model’s classification performance across both classes.

TABLE 3.5: Confusion matrix for Categorical Naïve Bayes (binary classification). Rows represent true labels; columns represent predicted labels.

	Predicted: Low (0)	Predicted: High (1)
True: Low (0)	1778	344
True: High (1)	672	1455

3.2.3 Decision tree - Binary classification

The Decision Tree classifier was optimized using Grid Search and Randomized Search with 5-fold cross-validation. Grid Search yielded slightly better cross-validation performance (CV score: 0.774 vs 0.771) and was selected for its more systematic exploration. The optimal configuration (entropy criterion, max_depth = 20, $\alpha_{ccp} = 0.0005$) achieved test accuracy of 0.772 with balanced performance: precision 0.73/0.83 and recall 0.86/0.69 for Low/High Interaction respectively (macro F1: 0.77, AUC: 0.85). This represents the best overall performance among the three classifiers.

The decision tree structure (Figure 3.4) highlights that user interaction is primarily driven by the strategic nature of the game. Strategy games, especially those with expansions, variants, and focused player counts, are more likely to achieve high user interaction, while family-oriented games with limited extensibility tend to exhibit lower participation. This confirms that user interaction emerges from a combination of categorical and structural features rather than a single dominant factor.

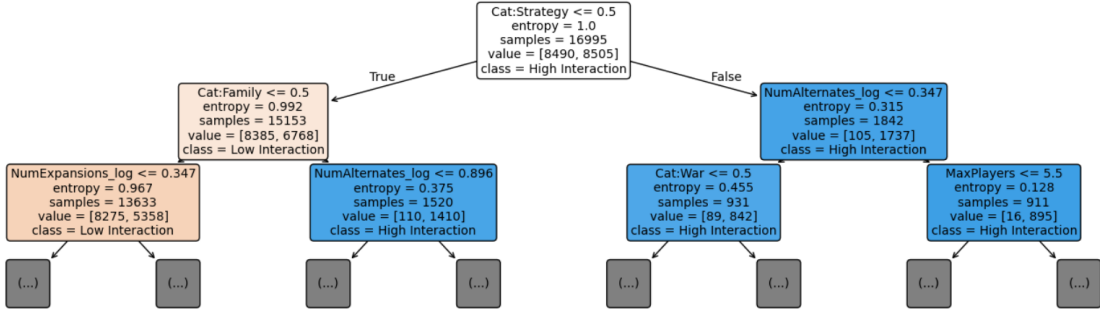


FIGURE 3.4: Decision Tree structure for binary classification, showing the hierarchical split conditions for predicting *HighInteraction*.

3.2.4 Summary

User interaction prediction proved challenging but tractable. All three models achieved competitive performance, with Decision Tree leading (highest accuracy and AUC), followed by Categorical Naïve Bayes and K-NN. Notably, no single feature dominates prediction—instead, user engagement emerges from complex, non-linear interactions between structural game characteristics and categorical attributes, highlighting the multifaceted nature of board game appeal.

4. Regression

Different regression techniques were applied to the dataset using *GameWeight* as the target variable. Univariate and multiple linear models, as well as non-linear regressors, were evaluated to assess the relationship between game complexity and selected explanatory features.

4.1 Univariate regression

Univariate regression analysis was first conducted to examine the relationship between game complexity (*GameWeight*) and recommended age (*ComAgeRec*). The predictor variable was chosen for its clear interpretability and a moderately strong monotonic association with the target variable (Spearman’s $\rho \approx 0.61$).

Table 4.1 reports the results obtained using different linear regression models, including estimated coefficients, intercepts, coefficient of determination (R^2), Mean Squared Error (MSE), and Mean Absolute Error (MAE).

TABLE 4.1: Univariate regression results for *GameWeight* prediction.

Algorithm	Coefficient	Intercept	R^2	MSE	MAE
Linear Regression	0.163	0.392	0.361	0.412	0.493
Ridge Regression	0.163	0.392	0.361	0.412	0.493
LASSO Regression ($\alpha = 0.01$)	0.162	0.404	0.361	0.412	0.493

All three linear models yield identical or nearly identical performance metrics ($R^2 = 0.361$, $MSE = 0.412$, $MAE = 0.493$), indicating a stable linear relationship between the predictor and the target variable. The absence of performance gains from regularization suggests limited multicollinearity and a well-conditioned univariate setting.

4.2 Multiple regression - linear

To capture additional explanatory power, a multiple linear regression framework was adopted by incorporating *PlaytimeComAvg* and *Rating*¹ as additional predictors. These variables exhibit substantial correlations with the target variable ($\rho \approx 0.69$ and $\rho \approx 0.51$, respectively).

As reported in Table 4.2, the inclusion of these features leads to a marked improvement in model performance, with R^2 increasing from 0.361 to 0.552, indicating that approximately 55.2% of the variance in *GameWeight* is explained by the linear model.

TABLE 4.2: Multiple linear regression results for *GameWeight* prediction.

Algorithm	Coefficients			Intercept	R^2	MSE	MAE
	<i>ComAgeRec</i>	<i>PlaytimeComAvg</i>	<i>Rating</i>				
Linear Regression	0.089	0.004	0.306	0.298	0.552	0.288	0.415
LASSO ($\alpha = 0.01$)	0.088	0.004	0.287	0.337	0.552	0.289	0.416
Ridge Regression	0.089	0.004	0.306	0.298	0.552	0.288	0.415

Among the predictors, *Rating* exhibits the strongest effect ($\beta \approx 0.306$), followed by *ComAgeRec* ($\beta \approx 0.089$), while *PlaytimeComAvg* contributes only marginally ($\beta \approx 0.004$). As observed in the univariate case, regularization techniques do not provide performance improvements over standard linear regression, yielding nearly identical evaluation metrics.

4.3 Multiple regression - non-linear

To assess the presence of non-linear relationships, Decision Tree and K-Nearest Neighbors (KNN) regressors were evaluated. Hyperparameter tuning was performed using Grid Search for the Decision Tree regressor (`max_depth = 10`, `min_samples_leaf = 4`, `min_samples_split = 10`), while the KNN regressor was optimized via cross-validation with $k = 20$ neighbors.

As shown in Table 4.3 the Decision Tree regressor achieves the best performance, explaining approximately 61.4% of the variance in game complexity ($R^2 = 0.614$). This represents a moderate but consistent improvement over linear models ($R^2 = 0.552$), indicating that the

¹*Rating* was encoded using an ordinal scale; details are provided in Section 1.4.2.

relationship between *ComAgeRec*, *PlaytimeComAvg*, *Rating*, and *GameWeight* involves non-linear interactions that cannot be fully captured through simple linear combinations.

TABLE 4.3: Multiple non-linear regression results for *GameWeight* prediction.

Algorithm	R^2	MSE	MAE
Decision Tree Regressor	0.614	0.248	0.378
KNN Regressor	0.604	0.255	0.383

5. Pattern Mining

To perform association analysis, continuous variables were discretized into categorical bins according to the scheme reported in Table 5.1. Binary genre indicators (*Cat:**) were included alongside the discretized features. The target variable *Rating*, encoded in Section 1.4.2, was also incorporated to enable predictive rule extraction.¹

TABLE 5.1: Discretization of continuous variables into categorical bins. Binning thresholds were determined empirically based on data distribution and interpretability criteria to ensure meaningful and balanced groupings.

Variable	Categories (Ranges)
<i>GameWeight</i>	Light (1–1.5), LightMedium (1.5–2.5), MediumHeavy (2.5–3.5), Heavy (>3.5)
<i>PlaytimeComAvg</i>	Short (0–30), Medium (30–60), Long (60–120), VeryLong (>120)
<i>YearPublished</i>	Pre1990, 1990–2004, 2005–2014, Post2015
<i>NumUserRatings</i>	Niche (0–56), Known (56–124), Popular (124–400), VeryPopular (>400)

5.1 Frequent pattern extraction

Frequent pattern extraction was performed using both Apriori and FP-Growth algorithms. An iterative approach was employed to determine the optimal support threshold, testing values from 1% to 30%.

As shown in Figure 5.1, frequent and closed itemsets were identical across all support values, indicating no redundancy in the discovered patterns.

¹In the context of association rule mining, the ordinal encoding values (1, 2, 3) are mapped to categorical labels (*LowRating*, *MediumRating*, *HighRating*).

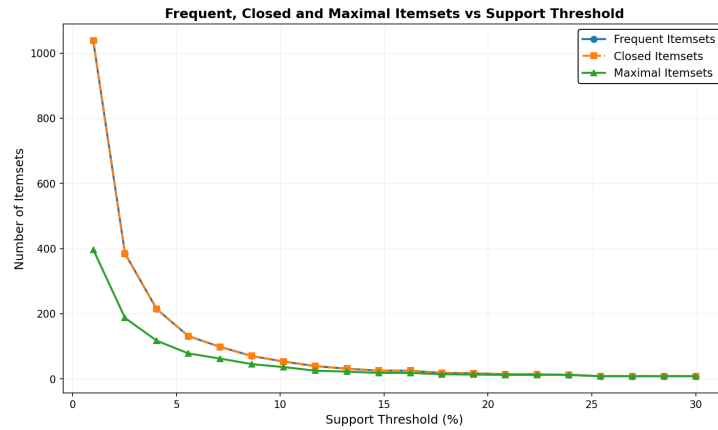


FIGURE 5.1: Comparison of frequent, closed, and maximal itemset cardinalities across support thresholds.

Based on the analysis, a support threshold of 10% was selected, yielding 59 frequent itemsets and 38 maximal itemsets. Both Apriori and FP-Growth produced identical results.

5.2 Association rules extraction

Association rules were extracted at varying confidence thresholds (10–100%) with the minimum support fixed at 10%. As shown in Figure 5.2, a threshold of 50% was selected at the elbow point, yielding 16 identical rules for both Apriori and FP-Growth. Table 5.2 presents the top 10 rules ranked by lift.

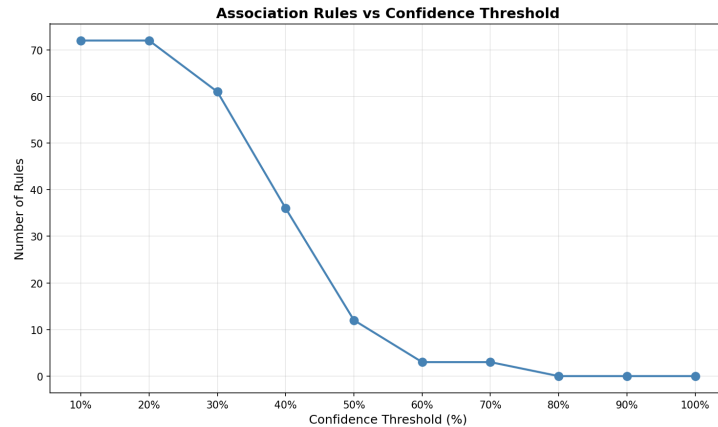


FIGURE 5.2: Variation in the number of association rules with respect to the confidence threshold.

TABLE 5.2: Top 10 association rules ranked by lift.

Antecedents	Consequents	Ant. Support	Cons. Support	Support	Confidence	Lift
(Short, LowRating)	(Light)	0.175	0.321	0.127	72.6%	2.26
(Light, 2005-2014)	(Short)	0.124	0.408	0.101	81.5%	2.00
(MediumRating, Light)	(Short)	0.126	0.408	0.101	79.6%	1.95
(2005-2014, Short)	(Light)	0.167	0.321	0.101	60.5%	1.88
(Light)	(Short)	0.321	0.408	0.240	74.8%	1.84
(Short)	(Light)	0.408	0.321	0.240	59.0%	1.84
(HighRating)	(Post2015)	0.233	0.321	0.136	58.6%	1.82
(Light, LowRating)	(Short)	0.176	0.408	0.127	71.9%	1.76
(LowRating)	(Light)	0.324	0.321	0.176	54.4%	1.69
(Light)	(LowRating)	0.321	0.324	0.176	54.9%	1.69

5.3 Target prediction

For predicting the target variable *Rating*, the analysis focused on rules with rating categories as consequents. Three rules predicting *LowRating* were identified (Table 5.3), while no rules with sufficient confidence were found for *HighRating* or *MediumRating* as direct consequents.

TABLE 5.3: Association rules predicting *LowRating*, ranked by lift.

Antecedents	Ant. Support	Cons. Support	Support	Confidence	Lift
Light	0.321	0.324	0.176	54.9%	1.69
(Short, Light)	0.240	0.324	0.127	52.8%	1.63
1990–2004	0.203	0.324	0.102	50.4%	1.55

The strongest rule, $Light \rightarrow LowRating$ (lift = 1.69), indicates that games with low complexity are 69% more likely to receive low ratings than expected by chance. One possible interpretation is that the BoardGameGeek community, primarily composed of hobbyist players, may prefer games offering greater strategic depth. When applied to the dataset, this rule identifies 32.1% of games, of which 54.9% actually have low ratings.

The second rule, $(Short, Light) \rightarrow LowRating$ (lift = 1.63), reinforces this pattern by combining low complexity with short playtime, suggesting that these characteristics may jointly contribute to lower ratings.

The third rule reveals a temporal association: games published between 1990 and 2004 are 55% more likely to receive low ratings (lift = 1.55). This could reflect evolving design standards, shifting community preferences, or reduced engagement with older titles.